National Institute of Open Schooling (NIOS)
Senior Secondary Course
Lesson – 29: Applications of Derivatives
Worksheet -29

1. Find the rate of change of the total surface area of a cylinder of radius (r) and height (h), when the radius varies

2. The radius of a circle is increasing uniformly at the rate of 5 cm/sec. Find the rate at which the area of the circle is increasing when the radius is 10 cm.

3. Discuss the applicable of Rolle's theorem on the function

$$f(x)=\begin{cases} x^2+1, \text{ when } 0\le x\le 1 \\ 3-x, \text{when } 1\le x\le 2 \end{cases}$$

4. Find the point on the curve $y=\dfrac{x}{1+x^2}$, where the tangent to the curve has the greatest slope.

5. If x is a real, find the minimum value of $x^2-8x+17$

6. Find the local maxima and local minimum of the function $f(x)=\sin x-\cos x, 0<x<2\pi$. Also find the local maxima and local minimum value.

7. PQ is the diameter of a circle and R is any point on the circle. Show that the area of triangle PQR is maximum, when it is on isosceles triangle.

8. Show that the local maximum value of $x+\dfrac{1}{x}$ is less than local minimum value.

9. Find the slope of tangent to the curve $x=a(\theta-\sin\theta)$, $y=a(1-\cos\theta)$ at $\theta=\dfrac{\pi}{2}$

10. Find the equation of tangent to the curve $y=x^2-2x+7$, which is perpendicular to the line $5y-15x=13$